Catalytic testing and kinetic measurements of GaNi Supported catalytically active liquid metal solutions (SCALMS) for selective oligomerization of ethylene

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The selective oligomerization of short olefins provides a direct route to a wide range of chemical products of substantial interest such as butenes, octenes and other industrial products.[1] Typical heterogeneous catalysts used in the oligomerization reaction do not provide high stability, selectivity and yield toward the desired linear products, compared to homogeneous catalysts[2]. Recently, *Supported catalytically active liquid metal solutions* (SCALMS) were developed as an alternative to facilitate dynamic single atom catalysis.[3] SCALMS are a new class of materials where small amounts of an active metal (e.g. Pt, Pd, Ni) are integrated in a low melting metal (e.g. Ga) matrix, resulting in liquid alloys at reaction conditions. [4]

In this contribution, we demonstrate the synthesis, catalytic testing (See **Figure 1**) and kinetic measurements of the Ga-Ni SCALMS materials for the oligomerization reaction.

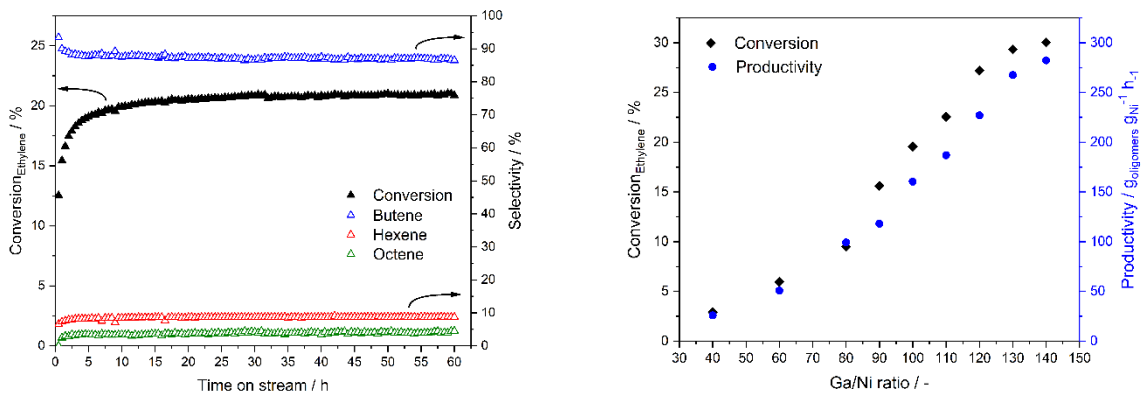

Figure 1: Conversion and selectivity vs. time on stream for oligomerization of ethylene using SCALMS catalyst with Ga/Ni = 100 (left). Conversion and productivity vs. Ga/Ni ratio for synthesised Ga_xNi-SCALMS catalysts tested in the oligomerization of ethylene (right).

(**Figure 1 left**) shows the results of a long term stability experiment using 1 g of the Ga/Ni= 100 SCALMS catalyst for ethylene oligomerization. The catalyst was stable

throughout the experiment with high selectivity towards dimers (C_4). This durability is attributed to the highly dynamic nature of the SCALMS liquid metal/gas interface. The long-term robustness is a clear advantage compared to literature known heterogeneous systems.

(**Figure 1 right**) shows the influence of the Ga/Ni ratio on the conversion and productivity of the catalyst. The higher Ga/Ni ratio resulted in higher conversion and productivity, highlighting the SCALMS effect. The higher conversion with higher Ga/Ni ratio is due to the mobile dispersed Ni atoms within the Ga droplets.

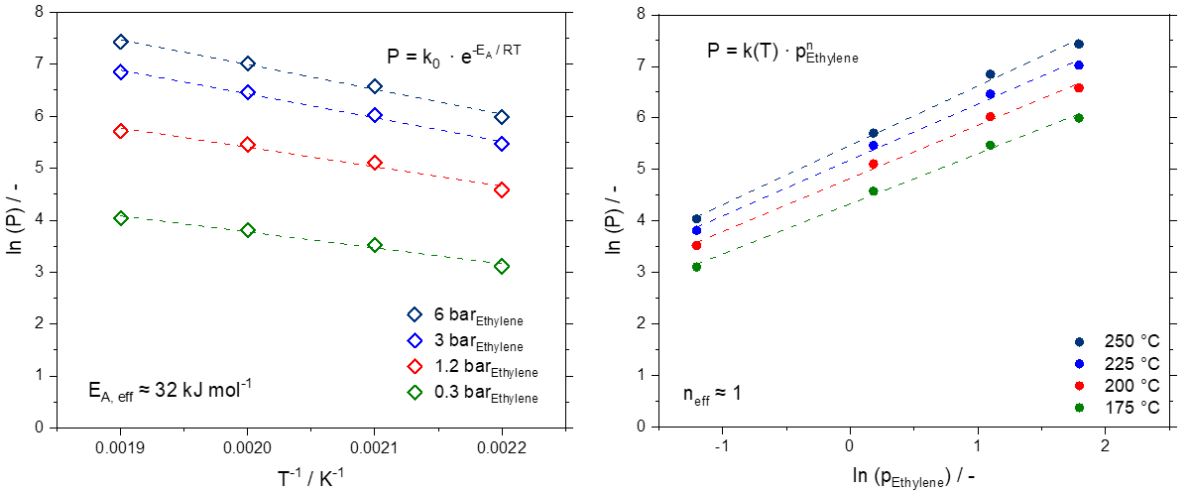

Figure 2: Arrhenius-plot with logarithmic P over inversed temperature (left). Plot of the logarithmic P over the logarithmic partial pressure for partial pressure variation (0.3 to 6 bar) at different temperatures (right) for a $Ga_{90}Ni$ SCALMS catalyst.

(**Figure 2)** shows the results of the kinetic measurements conducted with a Ga/Ni = 90. These data form the basis for process simulation studies which are currently investigated to pinpoint heat integration benefits of high temperature oligomerization.

References

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